Responsibility Convergence in Organizational Investigations

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Abstract

Organizational investigations such as root cause analysis (RCA) are intended to identify systemic causes of failure and support organizational learning. Yet across regulated and safety-critical environments, investigative conclusions recurrently stabilize responsibility at identifiable individuals despite formal commitments to system-oriented explanation. This persistent actor-centered pattern presents a theoretical puzzle: why do investigative systems designed to diffuse responsibility across sociotechnical structures so often concentrate accountability at particular organizational actors?

This paper develops a structural account of this recurrent stabilization dynamic, termed **responsibility convergence** in organizational investigations. Responsibility convergence refers to the patterned concentration of accountability at identifiable actors when investigative processes translate causal ambiguity into administratively actionable conclusions. The framework specifies how convergence arises from the sequential interaction of three mechanisms — procedural ambiguity, attributional closure pressure, and accountability legibility bias — operating within investigative reasoning under institutional constraints.

Under conditions of underdetermined causal reconstruction, organizations face institutional pressure to stabilize responsibility assignments. Because identifiable individuals provide the most interpretable and actionable attribution anchors within

accountability infrastructures, attribution systematically concentrates on actors rather than distributed structural conditions. Four theoretical propositions specify how variation in procedural ambiguity, closure pressure, and legibility bias conditions the strength and distribution of convergence across investigative contexts, and boundary conditions identify when convergence dynamics weaken.

Situated within a broader multi-stage responsibility lifecycle, responsibility convergence specifies the investigative stage at which causal ambiguity is translated into stabilized accountability assignments — complementing anticipated accountability convergence (AAC), which structures responsibility expectations prior to action; cognitive friction, which identifies how procedural texts embed localized judgment demands during execution; and AI-mediated accountability frameworks, which examine how computational representations further compress and stabilize responsibility assignments in post-investigative contexts. By situating investigative convergence within broader organizational responsibility alignment dynamics, the paper clarifies how RCA outcomes reflect not deviation from systemic ideals but the structural operation of accountability infrastructures — and outlines implications for organizational learning and accountability design.

Keywords: responsibility convergence, root cause analysis, accountability, organizational investigations, closure pressure, legibility bias, regulated environments

1 Introduction

Organizational investigations are central mechanisms through which institutions interpret failures and stabilize responsibility attribution. In safety-critical and highly regulated domains, adverse events rarely have a single cause but emerge from interactions among technologies, procedures, and human action. Investigative frameworks such as root cause analysis (RCA) are designed to diffuse responsibility across sociotechnical systems and identify contributing factors across organizational levels rather than individual culpability (Reason, 1990). These approaches reflect a long-standing recognition in safety science and organizational theory that failures in complex systems are distributed phenomena

requiring systemic explanation (Perrow, 1984; Vaughan, 1996).

Beyond their analytic function, investigations also play a constitutive organizational role. They are the processes through which ambiguous events are rendered intelligible, accountability is assigned, and lessons for future prevention are articulated. Across domains such as safety incidents, compliance failures, and operational disruptions, investigations operate as formalized episodes of organizational sensemaking in which causally underdetermined situations must be translated into administratively actionable accounts that stabilize responsibility (Weick, 1995). The tension between distributed causality and determinate accountability makes investigative settings revealing sites for examining responsibility stabilization in practice.

However, across regulated and safety-critical environments, investigative structures tend to stabilize responsibility at identifiable actors such as operators, supervisors, or local decision makers, even when formal frameworks emphasize systemic causation. Accident inquiries, audit reviews, and compliance investigations recurrently channel corrective actions centered on retraining, procedural adherence, or disciplinary measures directed at specific individuals or roles (Dekker, 2007; Power, 1997). In pharmaceutical GMP environments, for instance, deviation investigations and CAPA processes formally require systemic root cause identification — yet corrective actions disproportionately target operator retraining, procedural re-qualification, or individual performance review rather than upstream structural conditions. This pattern persists despite explicit commitments to system-level explanation, suggesting that actor-centered attribution reflects structural properties of investigative reasoning rather than merely inadequate methodology or intentional blame.

The recurrent stabilization of accountability at identifiable actors therefore poses a theoretical puzzle: why do investigative systems designed to diffuse responsibility across sociotechnical structures so often stabilize attribution at particular organizational actors? Existing accounts offer partial explanations. Attribution theory identifies dispositional tendencies under uncertainty (Heider, 1958; Ross, 1977), but does not specify how organizational procedures and accountability infrastructures structure these tendencies at

the system level. Safety science has documented the gap between systemic investigation ideals and person-centered outcomes (Dekker, 2007; Reason, 1997), but has not fully theorized the structural mechanisms through which this gap is produced. Accountability research has shown that evaluation expectations shape justification processes (Tetlock, 1985; Lerner & Tetlock, 1999), but has not integrated these dynamics into a mechanism model of investigative attribution.

This paper addresses this puzzle by introducing **responsibility convergence** — the structural stabilization of accountability at identifiable actors when investigative processes, structured by accountability infrastructures, translate causal ambiguity into administratively actionable conclusions. Rather than reflecting intentional blame or individual disposition alone, convergence arises from structural properties of investigative reasoning and accountability infrastructures. The analysis develops a mechanism model in which procedural ambiguity, attributional closure pressure, and accountability legibility bias sequentially channel attribution toward actors who provide the most interpretable, documentable, and sanctionable anchors for organizational action.

Situating investigative convergence within broader responsibility dynamics, the paper contributes to organizational theory in three ways. First, it reframes blame concentration as a predictable structural stabilization of responsibility rather than a deviation from systemic ideals. Second, it integrates attribution theory with accountability and governance perspectives by specifying how responsibility becomes operationally actionable within organizational procedures under conditions of causal ambiguity. Third, it positions responsibility convergence as the investigative stage within a broader multi-stage responsibility lifecycle — complementing anticipated accountability convergence (AAC), which structures expectations of accountable action prior to execution; cognitive friction, which identifies how procedural texts embed localized judgment demands during execution; and AI-mediated accountability frameworks, which examine how computational representations further compress and stabilize responsibility assignments in post-investigative contexts. Each mechanism is analytically distinct yet operates within the same temporally extended accountability structure — together tracing responsibility structuring across the full arc

of regulated organizational work.

2 Responsibility Attribution in Organizational Investigations

Responsibility attribution in organizations has been examined across attribution theory, accountability research, and organizational blame studies (Heider, 1958; Weiner, 1985; Roberts, 1991). Attribution theory distinguishes between dispositional and situational explanations of events, while organizational research has shown that negative outcomes are often attributed to individuals under conditions of uncertainty or causal complexity (Staw, 1975; Ross, 1977). Accountability scholarship further demonstrates that expectations of evaluation shape decision and justification processes (Tetlock, 1985; Lerner & Tetlock, 1999). However, these perspectives primarily explain attribution as a cognitive or evaluative process rather than as an outcome structured by investigative systems — leaving undertheorized how organizational procedures and accountability infrastructures shape attribution outcomes at the system level.

In organizational investigations, procedures such as RCA integrate attributional reasoning with procedural governance (Rooney & van den Heuvel, 2004; Andersen & Fagerhaug, 2006). Investigators must construct causal narratives linking events, actions, and outcomes within procedurally structured investigative frameworks. Yet organizational failures rarely yield complete causal reconstruction, leaving causal relations structurally indeterminate (Perrow, 1984; Vaughan, 1996). In pharmaceutical GMP environments, for instance, a single deviation event may involve equipment performance, environmental monitoring data, operator documentation, and upstream material variability — each representing a plausible causal pathway, none definitively established by available evidence. Investigators must nonetheless produce a determinate root cause assignment to satisfy regulatory documentation requirements and enable corrective action planning, even when the evidentiary basis for that assignment remains partial.

This structural indeterminacy produces a closure requirement for accountability sta-

bilization. Investigations cannot remain indefinitely ambiguous; they must stabilize conclusions to enable corrective action, regulatory documentation, and organizational learning processes (Dekker, 2007; Reason, 1997). The requirement to close investigations within defined timeframes — a standard feature of GMP deviation management systems — further intensifies closure pressure by converting an open-ended interpretive process into a time-bounded administrative task.

Within this context, responsibility convergence can be understood as a stabilization dynamic: attribution settles on targets that satisfy interpretive clarity and administrative actionability under ambiguity. Because organizational accountability expectations typically orient toward identifiable actors (Roberts, 1991; Power, 1997), investigative closure tends to settle on individuals even when causation is distributed. RCA procedures formalize this convergence dynamic by requiring determinate causal attribution and corrective assignment even when causal reconstruction remains distributed or indeterminate — structurally privileging attribution targets that can be clearly recorded, acted upon, and closed within organizational systems.

3 Mechanisms of Responsibility Convergence

The analysis identifies three interacting mechanisms that generate responsibility convergence in organizational investigations: procedural ambiguity, attributional closure pressure, and accountability legibility bias. These mechanisms operate sequentially and interact: procedural ambiguity structurally intensifies closure pressure; closure pressure in turn activates and amplifies legibility bias; legibility bias ultimately directs attribution toward identifiable actors. Responsibility convergence therefore emerges structurally from investigative reasoning under accountability constraints, without requiring intentional blame or deliberate individual attribution.

Procedural Ambiguity

Procedural ambiguity refers to indeterminacy in reconstructing causal sequences and accountability relations (Langley, 1999; Vaughan, 1996; Weick, 1995). Investigations rely on retrospective interpretation, incomplete documentation, and contested narratives. When causal chains cannot be definitively established, multiple attribution pathways remain viable and responsibility assignments remain indeterminate, generating pressure toward attributional convergence.

The interpretive indeterminacy introduced by procedural ambiguity does not remain neutral within investigative processes. Because investigations are conducted under expectations of explanation and resolution, ambiguity generates pressure for investigative closure: the need to transform open-ended interpretive space into determinate accounts that can support organizational action. Ambiguity thus structurally amplifies the demand for closure by expanding the range of plausible interpretations that must ultimately be resolved into accountable conclusions.

In pharmaceutical GMP deviation investigations, procedural ambiguity is structurally endemic. A contamination event, an out-of-specification result, or a documentation discrepancy may involve equipment history, environmental monitoring trends, operator actions, and upstream process variability — each representing a plausible causal pathway that is individually incomplete and collectively indeterminate. Investigators working within defined documentation templates and timeframe requirements must nonetheless resolve this interpretive field into a single root cause assignment. The ambiguity is not a failure of data collection but a structural feature of retrospective causal reconstruction in complex sociotechnical systems.

Attributional Closure Pressure

Investigative processes require determinate conclusions to support corrective action, regulatory reporting, and organizational continuity (Power, 1997; March et al., 2000; Dekker, 2007). Attributional closure pressure arises from this requirement. As ambiguity persists, pressure increases to select an explanation that is defensible and actionable,

transforming open causal interpretation into determinate attribution outcomes.

As investigative processes move toward resolution, closure pressure structures the selection environment of attribution. Among multiple plausible causal accounts, investigators are driven to privilege explanations that can sustain organizational scrutiny and support actionable decisions. This narrows the interpretive field, channeling attribution toward determinate outcomes that can be justified, documented, and operationalized.

Closure pressure in GMP environments is institutionally embedded at multiple levels simultaneously. Regulatory requirements mandate documented root cause assignments within defined timeframes. Quality management systems structure investigation workflows with mandatory completion fields, approval routing, and effectiveness check scheduling — converting open-ended causal inquiry into a sequenced administrative process with defined closure criteria. Internal audit and management review processes further intensify closure pressure by treating open or unresolved investigations as compliance risks requiring escalation. Under these compounding institutional pressures, investigators are systematically driven toward selecting attribution targets that can be documented, approved, and closed.

Accountability Legibility Bias

Accountability systems orient toward identifiable agents as attribution targets (Latour, 1987; Roberts, 2009). This orientation produces an **accountability legibility bias**: responsibility is more readily assigned to entities that are traceable, documentable, and sanctionable within organizational records and governance processes. Individual actions typically satisfy these criteria because they appear discrete and normatively evaluable, whereas systemic causes remain diffuse, distributed, and structurally embedded.

In GMP quality systems, corrective actions assigned to individual operators, analysts, or supervisors are straightforwardly documentable, trainable, and verifiable through effectiveness checks — making them structurally more legible than corrective actions targeting documentation architecture, workflow design, or resource allocation, which require cross-functional coordination, system validation, or procedural reengineering. Responsibility convergence thus reflects not merely a tendency toward individual blame but

the structural requirement that accountability systems render causality administratively actionable under evidentiary constraint.

Convergence Dynamics

These three mechanisms interact to produce responsibility convergence as a structural outcome of investigative reasoning. Procedural ambiguity expands the interpretive field and generates demand for closure; closure pressure narrows the viable interpretation set by privileging defensible and actionable accounts; legibility bias directs attribution toward identifiable actors who satisfy accountability system requirements for traceability and sanctionability. Responsibility convergence therefore does not require intentional blame, individual failure, or inadequate investigative methodology — it emerges structurally from the interaction of ambiguity, institutional pressure, and accountability infrastructure design.

4 Propositions

The convergence model yields four theoretical propositions specifying how variation in procedural ambiguity, closure pressure, and legibility bias conditions the strength and distribution of responsibility convergence across investigative contexts.

Proposition 1

Greater procedural ambiguity in causal reconstruction increases responsibility convergence toward identifiable actors.

As ambiguity in causal reconstruction expands, multiple attribution pathways remain viable and responsibility assignments remain indeterminate. Under such conditions, investigative processes face increasing demands to stabilize attribution. Because accountability systems orient toward identifiable agents as the most interpretable and actionable attribution targets, ambiguity intensifies the likelihood that responsibility ultimately converges on individuals rather than diffuse systemic causes. This proposition is non-obvious in an

important respect: it is not that ambiguity makes individual attribution more epistemically justified — it makes it more structurally available. In GMP deviation investigations, this dynamic is observable when investigations involving multiple plausible contributing factors nonetheless produce root cause assignments concentrated at the most proximal human action — not because systemic factors have been ruled out but because they remain insufficiently specified to anchor a documentable corrective action.

Proposition 2

Attributional closure pressure strengthens the effect of procedural ambiguity on convergence toward identifiable actors.

Ambiguity alone does not determine where attribution converges; it produces a field of competing explanations. Closure pressure, however, requires the selection of a determinate and defensible account to enable organizational action. As closure pressure intensifies, investigative reasoning is increasingly driven to resolve ambiguity through attribution targets that can support corrective decision-making, thereby amplifying convergence toward identifiable actors. What makes this proposition theoretically significant is the interaction effect: closure pressure does not independently produce convergence but amplifies the convergence tendency already generated by ambiguity. Organizations that impose tight investigative timeframes, mandatory root cause fields, and escalation triggers for open investigations are therefore structurally compounding ambiguity-driven convergence — intensifying actor-centered attribution through the procedural architecture of the investigation system itself.

Proposition 3

Accountability legibility bias mediates the relationship between attributional closure pressure and attribution to identifiable actors.

Closure pressure necessitates determinate attribution, but legibility determines which targets can satisfy this requirement. Accountability infrastructures privilege attribution targets that are traceable, documentable, and sanctionable within organizational records.

Because individuals most readily meet these criteria, closure pressure is translated into actor-centered attribution through legibility bias. This mediation dynamic clarifies why convergence stabilizes at individuals rather than at other potentially determinate targets such as equipment or environmental conditions: individuals are not merely identifiable but are embedded within organizational accountability structures — trainable, supervisable, and disciplinable — in ways that make them the most administratively complete attribution anchors available within quality and compliance systems.

Proposition 4

Higher responsibility convergence toward identifiable actors is negatively associated with systemic learning depth in post-investigation corrective action processes.

When responsibility converges on identifiable actors, corrective processes tend to target behavior, compliance, or local decision quality rather than underlying structural conditions. Actor-centered attribution directs investigative attention toward proximal actions, limiting examination of distributed system interactions underlying failure — reducing systemic learning depth even when formal learning processes exist and investigators are genuinely committed to systemic improvement.

This proposition has direct practical implications in GMP-regulated environments. When a deviation investigation concludes with operator retraining as the primary corrective action, the documentation architecture, workflow design, or resource constraints that shaped the operator's interpretive context remain unexamined and uncorrected. Subsequent deviations arising from the same structural conditions are therefore predictable — and the recurrence pattern itself becomes evidence of constrained organizational learning produced by convergence.

Boundary Conditions

Responsibility convergence is not inevitable. Convergence dynamics are likely to weaken in organizational contexts where accountability structures are explicitly distributed across collective units rather than oriented toward identifiable actors. In settings characterized

by shared liability regimes, strong systemic transparency, and reduced sanction intensity at the individual level, attribution may remain more diffuse and structurally anchored.

Convergence may also weaken where investigative protocols are explicitly designed to resist closure pressure — for instance, through provisional conclusion frameworks that allow investigations to remain open pending additional data, or through multi-disciplinary review structures that distribute interpretive authority across functional boundaries.

A third boundary condition concerns the distribution of sanctionability across organizational levels. Where override authority, resource allocation decisions, and system design choices made at managerial levels are themselves subject to formal accountability review — rather than being treated as background conditions exempt from investigative scrutiny — legibility bias may extend upward in the organizational hierarchy rather than concentrating exclusively at operational actors. In such contexts, systemic causes become more legible as attribution targets because the actors responsible for structural conditions are themselves sanctionable within the investigative architecture.

5 Theoretical Contributions

This paper shows that blame concentration in organizational investigations is structurally produced — not a deviation from investigative ideals but a predictable outcome of accountability infrastructure design. Three theoretical contributions follow.

First, it reframes blame concentration as a structural convergence process. Prior accounts have emphasized dispositional tendencies — the fundamental attribution error, hindsight bias, or motivated reasoning — as primary explanations for person-centered attribution in investigations (Ross, 1977; Dekker, 2007). While these dynamics are real, they do not explain why convergence occurs so consistently across diverse organizational contexts, investigative methodologies, and investigators with genuine commitments to systemic explanation. Responsibility convergence reframes this consistency as a structural outcome: actor-centered attribution arises from the interaction of procedural ambiguity, closure pressure, and legibility bias within investigative systems. If convergence is structural

rather than dispositional, efforts to improve investigative quality through training or awareness alone will remain insufficient without addressing the procedural architecture and investigative conditions that generate convergence.

Second, the framework integrates attribution theory with accountability and governance literatures. Attribution research has identified tendencies toward dispositional explanation under uncertainty but has treated attribution primarily as a psychological process (Heider, 1958; Weiner, 1985). Accountability scholarship has shown that evaluation expectations shape justification processes but has not specified how these expectations become embedded in investigative procedures (Tetlock, 1985; Lerner & Tetlock, 1999). Responsibility convergence integrates these streams by showing how dispositional tendencies are amplified and directed by procedural governance structures — specifically, how closure requirements and legibility constraints within accountability infrastructures translate ambiguity-driven attribution pressure into determinate, actor-centered outcomes.

Third, it positions responsibility convergence as the investigative stage within a broader multi-stage responsibility lifecycle. Anticipated accountability convergence (AAC) specifies how responsibility expectations become aligned across procedural, documentary, and evaluative infrastructures prior to action. Cognitive friction identifies how procedural texts embed localized judgment demands during execution. Responsibility convergence specifies how causal ambiguity is translated into stabilized accountability assignments through investigative closure following events. AI-mediated accountability frameworks examine how computational representations further compress and stabilize responsibility assignments in post-investigative contexts. Each mechanism is analytically distinct yet operates within the same temporally extended accountability structure — and responsibility convergence clarifies the investigative stage at which distributed causality is transformed into administratively actionable attribution.

6 Implications for Organizational Learning and Accountability Design

Responsibility convergence has direct implications for both organizational learning and the design of accountability infrastructures in regulated environments.

Organizational learning deficits. When investigations concentrate responsibility on individuals, systemic vulnerabilities may be underdiagnosed and corrective actions may emphasize retraining or discipline rather than structural redesign. In GMP-regulated pharmaceutical environments, this dynamic is observable in deviation recurrence patterns: when root cause assignments consistently target operator behavior or procedural non-adherence, the same deviation categories reappear across investigation cycles because upstream documentation architecture, workflow design, or resource constraints remain uncorrected. Responsibility convergence thus produces constrained organizational learning that accumulates across investigation cycles, eroding organizational resilience over time.

Intervention design. Mitigating convergence requires interventions mapped to its underlying structural mechanisms. Three intervention directions follow from the mechanism model.

Reducing procedural ambiguity through richer data capture and temporal reconstruction tools may limit the interpretive indeterminacy that generates closure demand. In practice, this means investing in documentation systems that preserve causal evidence across the full event sequence — including environmental monitoring trends, equipment performance histories, and workflow interaction logs — rather than relying on retrospective operator accounts as the primary evidentiary basis for root cause reconstruction.

Designing investigative protocols that tolerate provisional conclusions may reduce closure pressure by decoupling administrative completion requirements from causal certainty requirements. Where regulatory frameworks permit, separating immediate corrective containment actions from longer-term root cause investigation timelines allows causal reconstruction to remain open while organizational continuity is maintained.

Enhancing the representation of systemic causal pathways within investigative docu-

mentation templates may counteract legibility bias by making distributed causes more interpretable and actionable within organizational conditions. This means designing corrective action frameworks that include structural categories — documentation architecture, workflow design, resource allocation, coordination structure — alongside individual behavior categories, so that systemic attribution targets become as administratively legible as actor-centered ones.

Accountability design implications. These interventions aim not to eliminate individual responsibility but to rebalance responsibility across organizational levels. Responsibility convergence is not eliminated by good intentions or investigator training alone — it is reproduced by the structural properties of investigative architectures. Redesigning those architectures to distribute legibility, reduce closure pressure, and expand the temporal scope of causal reconstruction is therefore a prerequisite for investigative systems that genuinely support systemic learning.

7 Discussion

Investigations stabilize responsibility by translating ambiguous causal structures into determinate accountability assignments under institutional constraints. In regulated environments, where actions must be documented, justified, and auditable, attributional legibility becomes a central organizing principle of investigative reasoning. Responsibility convergence therefore reflects how investigative closure processes align with organizational tendencies to orient accountability toward identifiable actors — and clarifies why this alignment is structurally reproduced across organizational contexts, investigative methodologies, and investigators with genuine commitments to systemic explanation.

Although developed in the context of root cause analysis, responsibility convergence is not unique to formal investigations. Similar convergence dynamics arise whenever organizations must translate causally underdetermined events into administratively actionable accounts, including performance evaluation, audit and compliance processes, and post-hoc decision justification. In pharmaceutical GMP environments, this generalization

is visible across multiple accountability contexts simultaneously: deviation investigations, CAPA effectiveness reviews, management review processes, and regulatory inspection responses all exhibit convergence dynamics, despite operating under different procedural frameworks. The same underlying pressures recur — ambiguous causal reconstruction generates demands for closure, while accountability infrastructures privilege attribution targets that are traceable, recordable, and sanctionable.

Responsibility convergence thus reveals how organizations transform distributed causality into administratively actionable attribution. Investigative conclusions represent stabilized points of responsibility alignment under uncertainty — points at which the tension between distributed causality and determinate accountability is resolved, provisionally and institutionally, in favor of the latter. When closure settles on an individual actor, the systemic pathways that remained interpretively viable prior to closure become organizationally invisible — not because they were ruled out but because the investigative process did not require their resolution. Subsequent corrective planning, effectiveness review, and management oversight all operate within the attribution frame established at closure, systematically narrowing the organizational field of view over successive investigation cycles (Weick, 1995; March et al., 2000).

Situated within the broader responsibility lifecycle, responsibility convergence specifies the investigative stage at which distributed causality is transformed into stabilized accountability assignments. Whereas cognitive friction identifies how procedural structures pre-configure localized judgment during task execution, responsibility convergence explains how investigative infrastructures subsequently stabilize responsibility attribution after events. And whereas AAC explains how responsibility expectations become aligned across organizational layers prior to action, responsibility convergence specifies how those anticipatory alignments interact with post-event investigative closure to produce determinate attribution outcomes.

8 Conclusion

Responsibility in organizational investigations systematically converges on identifiable actors. This paper introduces responsibility convergence as a structural explanation for this pattern — showing that blame concentration arises not from intentional attribution or investigator failure, but from the structural interaction of procedural ambiguity, attributional closure pressure, and accountability legibility bias within investigative systems.

The central contribution is not simply to observe that investigations produce person-centered outcomes — a pattern well documented across safety science and organizational research — but to specify the structural mechanisms through which this pattern is generated, the propositions governing how variation in those mechanisms conditions convergence strength, and the boundary conditions under which convergence dynamics weaken or become redistributed across organizational levels. Responsibility convergence thereby provides an analytically tractable account of how distributed causality is transformed into administratively actionable attribution.

Situated within a broader responsibility lifecycle, responsibility convergence specifies the investigative stage at which causal ambiguity is translated into stabilized accountability assignments. AAC explains how responsibility expectations become aligned across procedural, documentary, and evaluative infrastructures prior to action — shaping the interpretive and evidentiary conditions that investigators must work with after an event occurs. Cognitive friction identifies how procedural texts embed localized judgment demands during execution. Responsibility convergence then specifies how the evidentiary record produced through these prior stages is translated into stabilized attribution through investigative closure. AI-mediated accountability frameworks examine how computational representations further compress and stabilize responsibility assignments in post-investigative contexts. Each mechanism is analytically distinct yet operates within the same temporally extended accountability structure.

Recognizing convergence dynamics has practical implications for how regulated organizations design investigative systems, documentation architectures, and corrective action frameworks. If responsibility convergence is structurally produced rather than individ-

ually determined, improving investigative quality requires attending to the procedural architecture and accountability infrastructure conditions that generate convergence — not only to investigator training or methodology. Designing investigative systems that reduce procedural ambiguity, tolerate provisional conclusions, and enhance the legibility of systemic causal pathways represents a prerequisite for investigations that genuinely support organizational learning rather than merely producing administratively actionable attribution.

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